

Plan and Steps for ETL Process

Before applying any data transformation steps, it's essential to thoroughly analyze the files to understand their structure, content, and potential inconsistencies.

Tools used & Technologies

- Python 3.9+ for ETL scripting and analysis
- pandas for data cleaning and transformation
- Microsoft Excel for manual and easy cleanup
- PostgreSQL as the target data warehouse
- psycopg2 for PostgreSQL integration
- matplotlib & seaborn for visualizations

ETL Workflow Summary

Extract:

- Load raw offline data from CSV files (transactions and customer info)

Transform:

- Clean column names and formats
- Convert amounts to float and standardize dates
- Extract identifiers and periods from comment fields
- Infer transaction types (credit/debit) where missing
- etc.

Load:

- Load cleaned data into a PostgreSQL data warehouse using psycopg2

STEP 1: ANALYZE "Dummy Data Customer Info.csv"

Field	Problem Description
Duplicate Customer Records	Example: customer_id = 100156 appears three times with different emails and birthdates. Action: Decide on a deduplication rule, keep the most recent record.
Date of Birth Format Inconsistency	Formats example used 93-06-14 (YY-MM-DD), 14/6/1993 (D/M/YYYY), 6/14/93 (M/D/YY) 1995/10/25 (YYYY/MM/DD)

Resolved Issues in customer info

- Duplicate customer_id entries resolved by keeping the most recent birthdate.
- date_of_birth standardized to YYYY-MM-DD.
- Region names normalized (capitalized and stripped).
- No null values remain in required fields.

STEP 2: ANALYZE TXN Part 1.csv

Field	Problem Description
CUTOMER_ID	Misspelled – should be CUSTOMER_ID
AMOUNT	String with commas, needs conversion to float
CURRENCY	All NaN, but actual currency is in ACCOUNT column (could be RWF,USD and EURO)
TRANSACTION_TYPE	Appears to be transaction direction (e.g. 'C' = Credit, 'D' = Debit)
COMMENT	Contains embedded info (e.g. "IDNO 024910", period, etc.), needs parsing

Continuation: ANALYZE TXN Part 2.csv

Field	Problem Description
NAME, LAST NAME	Misaligned columns — these likely hold ACCOUNT, NAME, LASTNAME, CHANNEL
AMOUNT	In wrong column, includes description ("DISB LOAN JAN...") → needs extraction
ACCOUNT	Mostly missing

Resolved Issues in TXN data

- Rename and Fix Column Names
- Standardize both files to use the same column
- Normalize AMOUNT, Remove commas, extract the numeric part, and convert to float.
- Parse and Identifier and period From TRANSACTION_COMMENTS

Manual Data Fixes using Excel

TXN Part 1.csv

- Corrected the typo CUTOMER_ID to CUSTOMER_ID.
- Moved the currency under the correct column.
- Moved the account number under the correct column.
- Adjusted the NAME and LASTNAME to their correct columns.
- Created a column called TRANSACTION_TYPE and assigned values based on "D" (Debit) and "C" (Credit).

Continuation Manual Fixes

TXN Part 2.csv

- Deleted data inside the COMMENT column from lines 4 to 36.
- Copied data from the AMOUNT column (lines 2 to 36) and pasted it into the COMMENT column.
- Copied data from column E (lines 2 to 41) and exchanged it with column F (lines 2 to 41).

Conclusion

After applying all the outlined cleaning and transformation steps, the customer information will be fully standardized, with duplicate records resolved, date formats unified, and essential fields properly structured. The transaction data, originally fragmented across multiple files, will now be combined into a cohesive dataset with corrected column names, properly formatted amounts, and extracted key identifiers.